PROJECT STATUS REPORT

ClaudeCode Advancements

Research, tools, and systems for AI-assisted development

7,850	137,206	9	754
TESTS PASSING	LINES OF CODE	MODULES	COMMITS

33	368	18	31d
master tasks	intelligence findings	live hooks	project age

Session 122 · 2026-03-22

github.com/mpshields96/ClaudeCodeAdvancements

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EXECUTIVE SUMMARY

Executive Summary

ClaudeCodeAdvancements is a research and development project building validated tools, hooks, and systems for Claude Code users and AI-assisted development workflows. Launched 2026-02-19 with 122 development sessions completed. All five original research frontiers are production-complete with 7,850 automated tests across 9 modules. The project has expanded into 33 master-level tasks covering autonomous intelligence gathering, self-learning loops, academic research integration, and professional design output. Zero external Python dependencies.

Project Health

TESTS

7,850 / 7,850

198 suites — 100% pass rate

CODEBASE

137,206

Source: 55,238 Test: 81,968

Test-to-source ratio: 1.48:1

VELOCITY

754 commits

24.3/day · 6.2/session

31 days · 122 sessions

MODULES

9

141 source files · 199 test files

0 stubs · 0 syntax errors

MASTER TASKS

33

8 done · 19 active · 5 pending

INTELLIGENCE

368 findings

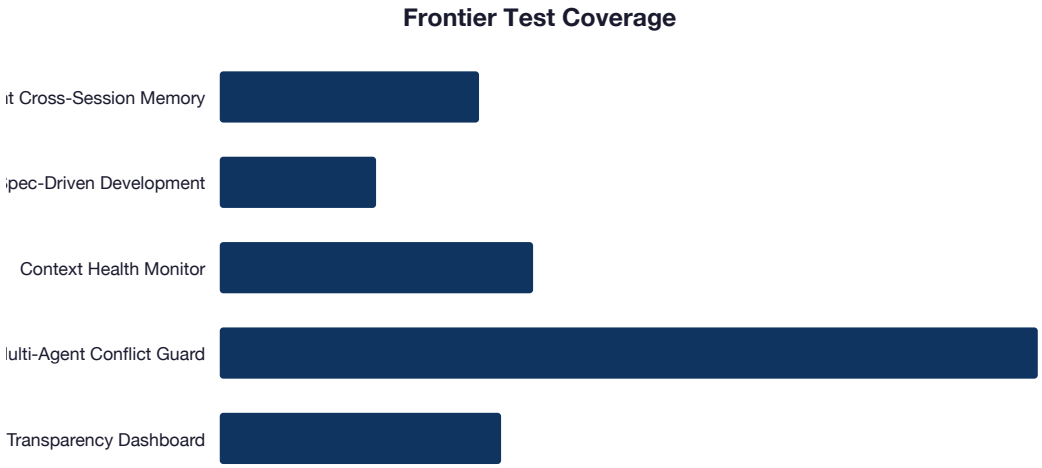
1242 papers · 18 live hooks

Zero external dependencies

FIVE FRONTIERS

Five Frontiers

The five core research areas that drive CCA's development, each validated through community intelligence, Anthropic research, and developer surveys.



1 Persistent Cross-Session Memory

Active CRITICAL

Every session starts from zero — this gives Claude a brain that persists.

TESTS: **340** LOC: **1,758**

2 Spec-Driven Development

Active HIGH

Unstructured prompting produces poor architecture. Enforce requirements > design > tasks > implement.

TESTS: **205** LOC: **1,186**

3 Context Health Monitor

Active HIGH

Context rot causes silent output degradation. Monitor, warn, and auto-handoff.

TESTS: **411** LOC: **2,681**

4 Multi-Agent Conflict Guard

Active HIGH

Parallel agents overwrite each other. Guard credentials, paths, and system modifications.

TESTS: **1073** LOC: **6,196**

5 Usage Transparency Dashboard

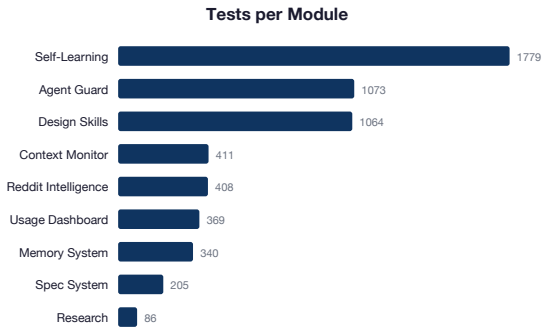
Active MEDIUM

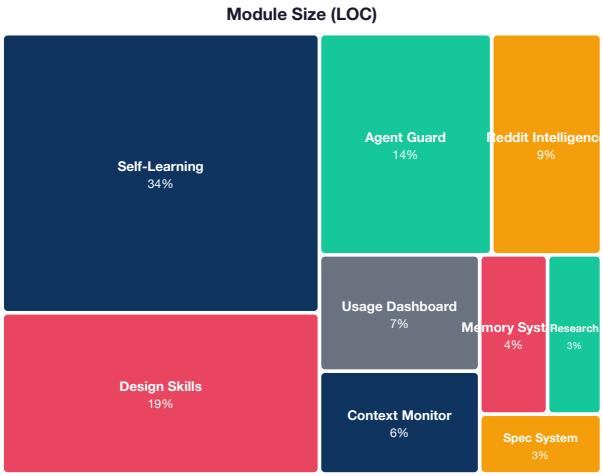
No real-time token/cost visibility. Counter, alerts, and structural completeness.

TESTS: **369** LOC: **3,030**

MODULE DEEP-DIVES

Module Deep-Dives





Memory System

Active

TESTS **340** LOC **1,758** FILES **4** `memory-system/`

Persistent cross-session memory. Captures decisions, preferences, and context across Claude Code sessions using hooks and MCP.

KEY COMPONENTS

- Capture hook (PostToolUse + Stop)
- CLI viewer (stats, search, list)
- 8-char hex ID schema
- MCP retrieval server
- Compaction-resistant handoff

WHY THIS MATTERS Means Claude remembers your preferences and past decisions across sessions — no more re-explaining context every time you start a new chat.

Spec System

Active

TESTS **205** LOC **1,186** FILES **4** `spec-system/`

Spec-driven development workflow. Enforces requirements > design > tasks > implement sequence with approval gates.

KEY COMPONENTS

- /spec:requirements interview scaffold
- /spec:tasks atomic decomposer
- PreToolUse spec validator hook
- /spec:design architecture generator
- /spec:design-review (4 expert personas)
- UserPromptSubmit auto-activation

WHY THIS MATTERS Forces Claude to plan before coding. You get a reviewable blueprint (requirements, design, tasks) instead of spaghetti code from vague prompts.

Context Monitor

Active

TESTS **411** LOC **2,681** FILES **10** `context-monitor/`

Context health monitoring with adaptive thresholds. Tracks token usage, warns at quality ceilings, blocks exit at critical levels.

KEY COMPONENTS

- PostToolUse token meter
- Stop hook auto-handoff
- Automatic session wrap trigger
- PreToolUse threshold alert
- Compact anchor (every 10 turns)
- Session pacer (2-3h autonomous runs)

WHY THIS MATTERS Warns you when Claude's memory is getting full so it doesn't silently start forgetting instructions mid-session. Prevents the 'why did it ignore my rules?' problem.

Agent Guard

Active

TESTS **1073** LOC **6,196** FILES **21** `agent-guard/`

Multi-agent safety, conflict prevention, and Senior Dev code review. Guards credentials, network ports, dangerous paths, and provides automated quality scoring on every write.

KEY COMPONENTS

- Mobile approver (ntfy.sh push)
- Network/port exposure guard
- Path validator (LIVE in hooks)
- File ownership manifest
- Bash command safety guard (LIVE in hooks)
- PR effort scorer (1-5 scale)
- False positive filter
- Tech debt tracker (trend analysis)
- Senior Dev orchestrator hook (LIVE)
- Credential extraction guard
- Content scanner (9 threat categories)
- Session guard (slop detection)
- Edit retry guard (LIVE in hooks)
- SATD detector (TODO/FIXME/HACK markers)
- Code quality scorer (0-100, A-F)
- Review classifier (6 categories)
- ADR reader (architecture decision discovery)

WHY THIS MATTERS Stops Claude from accidentally deleting files, exposing API keys, or breaking your computer when running autonomously overnight.

Usage Dashboard

Active

TESTS **369** LOC **3,030** FILES **7** `usage-dashboard/`

Token and cost transparency. CLI counter, OpenTelemetry receiver, cost alerts, and structural completeness checker.

KEY COMPONENTS

- Token counter CLI
- PreToolUse cost alert hook
- OTLP HTTP/JSON receiver
- /arewedone completeness checker

WHY THIS MATTERS Shows you exactly how many tokens and dollars each session costs, so you can spot waste and stay within your Max subscription limits.

Reddit Intelligence

Complete

TESTS **408** LOC **3,951** FILES **7** `reddit-intelligence/`

Community signal research. Discovers tools, patterns, and pain points from Reddit and GitHub to drive development priorities.

KEY COMPONENTS

- Reddit reader (posts + all comments)
- GitHub repo evaluator
- Subreddit profile registry (10 profiles)
- Autonomous scan pipeline
- Sandboxed repo tester

WHY THIS MATTERS Automatically finds the best new tools, tips, and techniques from Reddit so you don't have to scroll through hundreds of posts yourself.

Self-Learning

Complete

TESTS **1779** LOC **14,399** FILES **35** `self-learning/`

Cross-session improvement through structured observation, pattern detection, autonomous proposals, and research ROI tracking.

KEY COMPONENTS

- JSONL event journal
- Trace analyzer (4 detectors)
- Sentinel adaptive mutation
- Findings resurfacer
- Overnight detector (time-stratified analysis)
- Trade reflector (5 pattern detectors)
- Pattern reflector + strategy engine
- YoYo improvement loop
- Academic paper scanner
- Skillbook injection hook (LIVE)
- Research outcomes ROI tracker

WHY THIS MATTERS Makes your Kalshi bot smarter over time by analyzing what worked and what didn't. The bot learns from its own wins and losses instead of repeating mistakes.

Design Skills

Active

TESTS **1064** LOC **8,317** FILES **12** `design-skills/`

Professional visual output. Typst-based PDF reports, slides, HTML dashboards, SVG charts, website generation, and daily project snapshots.

KEY COMPONENTS

- Report generator (Typst pipeline)
- Dashboard generator (HTML)
- Website generator (landing + docs)
- Design guide + visual language
- Slide generator (16:9 PDF)
- Chart generator (SVG)
- Daily snapshot engine (metric diffs)

WHY THIS MATTERS Generates professional PDF reports and charts automatically — no manual formatting. One command gives you a polished status overview you could show anyone.

Research

Complete

TESTS **86** LOC **1,397** FILES **3** `research/`

R&D tools and experimental capabilities including iOS project generation and Xcode build wrapper.

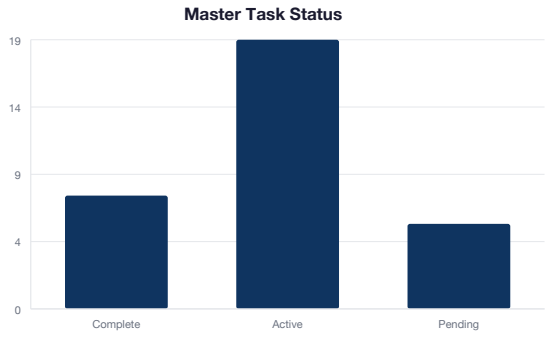
KEY COMPONENTS

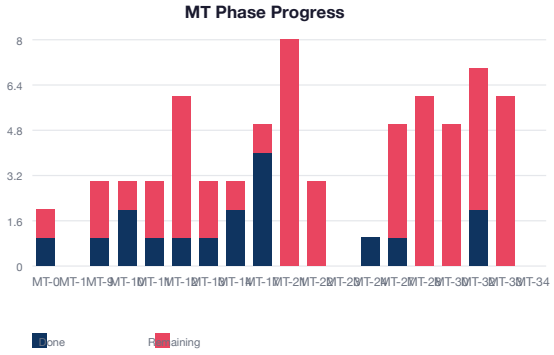
- iOS project generator (SwiftUI)
- Xcode build wrapper

WHY THIS MATTERS R&D sandbox for experimental features. Things get prototyped here before becoming real modules — keeps the main codebase clean.

MASTER TASKS

Master Tasks





33 tasks defined · **8 complete** · **19 active** · 5 pending · 1 blocked

Completed

8 tasks delivered

MT–2 Mermaid Architecture Diagrams in Spec System

Complete

Phase 1/1

COMPLETE (Session 19). Mermaid diagram requirement added to spec:design at lines 34 and 109.

MT–3 Virtual Design Team Plugin

Complete

Phase 1/1

COMPLETE (Session 21). `/spec:design-review` command created with 4 expert personas (UX, Security, Performance, Maintainability). Consolidated review with APPROVE/REVISE/REDESIGN verdict.

MT–4 Frontend Design Excellence via Design Vocabulary

Complete

Phase 1/1

COMPLETE (Session 21). Section 1b "Design References" added to spec:design — optional for UI features, includes reference UIs, design vocabulary, layout pattern, color constraints. Asks user once if no preferences specified.

MT-6 On-Demand Subreddit Scanner

Complete

Phase 1/1

COMPLETE (Session 27). Delivered:

DELIVERED

- profiles.py: 10 builtin subreddit profiles (4 domains: claude/trading/dev/res...
- --profile and --quick flags added to nuclear_fetcher.py

MT-7 Programmatic Trace Analysis for Self-Learning

Complete

Phase 1/1

COMPLETE (Session 26). Delivered:

DELIVERED

- self-learning/trace_analyzer.py: TranscriptEntry, TranscriptSession, RetryDet...

WHY THIS MATTERS Finds patterns in how Claude wastes tokens (retries, unnecessary reads) so sessions run faster and cheaper.

MT-20 Senior Developer Agent

Complete

Phase 0/9

COMPLETE (S71-S83). All 10 gaps closed + E2E validated 10/10 with real API (claude-haiku-4-5-20251001, S83).

REMAINING

- Phase 6: Hook output rewrite — natural language advice (S78) DONE
- Phase 7: /senior-review skill — APPROVE/CONDITIONAL/RETHINK verdicts (S78) DONE
- Phase 8: Interactive CLI chat — REPL + LLMClient + Anthropic API (S79-S80) DONE
- Phase 9: Architectural coherence — module structure, patterns, imports, rules (S78-S79...

NEXT: DONE. MT-20 is feature-complete. Converge with MT-21 Hivemind Phase 1.

WHY THIS MATTERS An automated code reviewer that catches bugs, tech debt, and design issues before they become problems. Like having a senior engineer on call 24/7.

MT-26 Financial Intelligence Engine

Complete

Phase 1/2

- Research COMPLETE (S104-105): MT26_FINANCIAL_INTEL_RESEARCH.md — papers, data sources, repos, 3-tier plan

WHY THIS MATTERS Financial intelligence tools for the Kalshi bot: regime detection, calibration analysis, order flow. Directly improves bet quality.

MT-29 **Cowork + Claude Pro Bridge Hivemind** Complete

Phase 1/1

Research COMPLETE (S114). Verdict: SKIP. Cowork adds no value over our hivemind for dev workflows. No Pro↔Code bridge exists. Local MCP bugs (GitHub #23424). See MT29_COWORK_RESEARCH.md.

Active

19 tasks in progress

MT-0 **Kalshi Bot Self-Learning Integration** In Progress

Phase 1/2

Phase 1 COMPLETE (Session 21). Trading domain schema built in CCA self-learning module:

DELIVERED

- get_trading_metrics() with PnL tracking, win rate, by-market-type and by-stra...
- Phase 2: Deploy to polymarket-bot (requires cross-project work)

REMAINING

- Phase 2: Deploy to polymarket-bot (requires cross-project work)

WHY THIS MATTERS The single most important task: making the Kalshi bot learn from its own trading results. Every dollar of improvement here compounds over time.

MT-1 **Maestro-Style Visual Grid UI** In Progress

MOSTLY SELF-RESOLVED (S96). Multiple tools now exist:

MT-9 Autonomous Cross-Subreddit Intelligence Gathering

In Progress

Phase 1/3

Phase 2 COMPLETE (Session 31). Delivered:

DELIVERED

- autonomous_scanner.py: ScanPrioritizer (staleness + yield + diversity scoring...
- execute_scan(): End-to-end pipeline — fetch → filter → dedup → classify → rec...
- ScanResult dataclass with to_dict/save_json for structured output.
- /cca-nuclear autonomous mode: auto-picks highest-priority sub via scanner.
- Phase 3: Supervised trial — run with Matthew monitoring for 3 sessions.

REMAINING

- Phase 3: Supervised trial — run with Matthew monitoring for 3 sessions.

WHY THIS MATTERS Automates Reddit scanning so high-value posts get flagged without you having to check manually.

MT-10 YoYo-Pattern Continuous Self-Learning and Self-Building

In Progress

Phase 2/3

Phase 1 COMPLETE (Session 28), QualityGate added (Session 29), E2E validated (Session 29).

DELIVERED

- improver.py: ImprovementProposal lifecycle, ImprovementStore (JSONL), Proposa...
- QualityGate: Geometric mean anti-gaming (Nash 1950 / sentrux pattern). Preven...
- Phase 2 COMPLETE (Sessions 42-44): 5/5 validation sessions, 36 transcripts an...
- Phase 3: Two tracks: (A) Graduate self-learning to Kalshi bot (requires cross...

REMAINING

- Phase 3: Two tracks: (A) Graduate self-learning to Kalshi bot (requires cross-project ...

WHY THIS MATTERS The 'get smarter each session' engine. Tracks what strategies work and surfaces the best ones automatically.

MT–11 Autonomous GitHub Repository Intelligence

In Progress

Phase 1/3

Phase 2 COMPLETE (Session 83). Delivered:

DELIVERED

- github_scanner.py: RepoMetadata (from GitHub API), RepoEvaluator (0-100 scori...
- fetch_trending(): GitHub search API with date filters to approximate trending...
- _build_trending_query(): configurable language, days, min_stars filters.
- TrendingScanner class: per-language scanning, trending history JSONL log, CCA...
- Phase 3: Wire trending scanner into MT-9 autonomous pipeline for scheduled runs.

REMAINING

- Phase 3: Wire trending scanner into MT-9 autonomous pipeline for scheduled runs.

MT–12 Academic Research Paper Integration

In Progress

Phase 1/6

Phase 4 COMPLETE (Session 101). Delivered:

DELIVERED

- self-learning/paper_scanner.py: Semantic Scholar + arXiv API integration, 7 C...
- self-learning/paper_digest.py: Kalshi/CCA digest generator with cross-chat br...
- Phase 1 (S38): Live API validated, first paper logged
- Phase 2: Ran across 4 domains, increased delay to 3s (429 rate limit)
- Phase 3 (S100): paper_digest.py built, expanded prediction/statistics queries...
- Phase 4 (S101): Expanded scans across prediction, statistics, trading_systems...
- Phase 5 (S102): Implemented confidence calibrator from ConfTuner (Li et al., ...
- Phase 6 (S102): Built hit_rate_tracker.py (32 tests). Computes APF from FINDI...

REMAINING

- Phase 2: Ran across 4 domains, increased delay to 3s (429 rate limit)

WHY THIS MATTERS Finds relevant academic papers (trading math, AI research) and summarizes them so you get the insights without reading 30-page PDFs.

MT–13 iOS App Development Capability

In Progress

Phase 1/3

Phase 2 COMPLETE (Session 49). Delivered:

DELIVERED

- ios_project_gen.py: Generates complete Xcode projects from CLI (SwiftUI + tes...
- xcode_build.py: Python xcodebuild wrapper (build, clean, test, scheme/simulat...
- Phase 3: Build first real app (Kalshi mobile dashboard) or install SwiftUI Ag...

REMAINING

- Phase 3: Build first real app (Kalshi mobile dashboard) or install SwiftUI Agent Skill

MT–14 Autonomous Re-Scanning of Previously Scanned Subreddits

In Progress

Phase 2/3

Phase 1 COMPLETE (Session 35). Delivered:

DELIVERED

- rescan_sub() method on AutonomousScanner: delta-scanning with timestamp filte...
- get_stale_subs() helper for identifying overdue subs
- Phase 2: Run rescan on a real stale sub, validate delta filtering works. Phas...

MT–17 UI/Design Excellence and Professional Report Generation

In Progress

Phase 4/5

Phase 1 COMPLETE (Session 41). Delivered:

DELIVERED

- design-skills/ module with CLAUDE.md, design-guide.md (color palette, typogra...
- design-skills/report_generator.py — CCADataCollector + ReportRenderer + CLI (...
- design-skills/templates/cca-report.typ — Professional status report template
- /cca-report slash command for one-command PDF generation
- Phase 2 COMPLETE (Session 47): slide_generator.py + /cca-slides command (25 t...
- Phase 3 COMPLETE (Session 48): dashboard_generator.py + /cca-dashboard comman...
- Phase 4 COMPLETE (Session 49): chart_generator.py — pure SVG chart generation...
- Phase 5: Website templates — landing pages, documentation sites

REMAINING

- Phase 5: Website templates - landing pages, documentation sites
- Phase 5: Website templates — landing pages, documentation sites

WHY THIS MATTERS One-command professional reports. Instead of explaining project status verbally, just generate a PDF.

MT–21 Hivemind Multi-Chat Coordination

In Progress

Phase 0/8

Infrastructure COMPLETE. Phase 1 validation IN PROGRESS (S86).

REMAINING

- Phase 1: Validated 2-chat (Desktop + 1 CLI), 3-5 sessions, must pass all gates
- Phase 2: Hardened 2-chat with complex tasks, 3-5 sessions
- Phase 3: 3-chat operation (Desktop + 2 CLI), only after Phase 2 gate passes

NEXT: 1-3 more validation tests, then Phase 1 gate assessment.

WHY THIS MATTERS Lets multiple Claude sessions work in parallel on different tasks — like having a small dev team instead of one assistant.

MT–22 Autonomous 1-Hour Loop

In Progress

Phase 0/3

GATE PASSED (S99). 3/3 supervised trials complete (S96 PASS, S97 PASS, S99 PASS). Approved for autonomous 1-hour segments. Phase 1 (S94), Phase 2 notification (S95), Phase 3 validation (S99) all COMPLETE. Base value: 9. Last touched: S99.

WHY THIS MATTERS Makes the whole system run without you. Sessions start, do work, wrap up, and hand off to the next session automatically.

MT–23 Mobile Remote Control v2

In Progress

EXTERNALLY RESOLVED (S112). Claude Code Channels shipped March 20, 2026.

REMAINING

- Install Telegram channel plugin and test with Kalshi bot session.

NEXT: Install Telegram channel plugin and test with Kalshi bot session.

MT–24 Visualization & Graphics Engine

In Progress

Phase 1/1

Phase 1 COMPLETE (S111) — `trading_chart.py` (5 SVG chart types: PnL curve, win rate, strategy matrix, drawdown, heatmap). 41 tests.

✓ 41 tests

MT–27 CCA Nuclear v2

In Progress

Phase 1/5

Phase 1-3 COMPLETE (S114), Phase 5 COMPLETE (S115).

DELIVERED

- Phase 1: APF analysis — "Other" category dropped from 124/335 (37%) to 0/335 ...
- Phase 2: classify_post HAY expansion — 28 new sentiment/opinion HAY keywords ...
- Phase 3: apf_checkpoint() function + scan report integration. 6 new tests.
- Phase 5 (S115): apf_session_tracker.py — per-session APF trend tracking. Appe...

REMAINING

- Phase 2: classify_post HAY expansion — 28 new sentiment/opinion HAY keywords (vibe cod...
- Phase 3: apf_checkpoint() function + scan report integration. 6 new tests.

WHY THIS MATTERS Smarter Reddit scanning with less noise. Better signal-to-noise means fewer irrelevant posts cluttering your review.

MT–28 Self-Learning v2

In Progress

Phase 0/6

COMPLETED (S111). All 6 phases done:

DELIVERED

- Phase 1: principle_registry.py (73 tests) — Laplace-scored principles by domain
- Phase 2: pattern_registry.py + detectors.py (42 tests) — Plugin detector arch...
- Phase 3: principle_transfer.py (34 tests) — Cross-domain affinity-based transfer
- Phase 4: outcome_feedback.py (16 tests) — Research outcomes → principle scoring
- Phase 5: predictive_recommender.py (40 tests) — Pre-session ranked recommenda...
- Phase 6: sentinel_bridge.py (30 tests) — SentinelMutator → principle registry...

REMAINING

- Phase 1: principle_registry.py (73 tests) — Laplace-scored principles by domain
- Phase 2: pattern_registry.py + detectors.py (42 tests) — Plugin detector architecture
- Phase 3: principle_transfer.py (34 tests) — Cross-domain affinity-based transfer
- Phase 4: outcome_feedback.py (16 tests) — Research outcomes → principle scoring
- Phase 5: predictive_recommender.py (40 tests) — Pre-session ranked recommendations

WHY THIS MATTERS Self-learning v2: the bot doesn't just track results, it evolves its own principles about what works. Adaptive strategy, not static rules.

MT-30 Session Daemon

In Progress

Phase 0/5

ALL PHASES COMPLETE (S111-S113).

DELIVERED

- [?] Phase 2 (S111): session_registry.py (60 tests) + tmux_manager.py (40 tests)
- [?] Phase 3 (S112): session_daemon.py — poll loop, health checking, spawn/restart...
- [?] Phase 4 (S113): Integration tests — lifecycle, peak transitions, crash recove...
- [?] Phase 5 (S113): Hardening — _mark_exhausted_sessions FAILED state fix, log ro...

REMAINING

- Phase 2: session_registry.py + tmux_manager.py (building blocks)
- Phase 3: session_daemon.py (poll loop, health checking, spawn/restart)
- Phase 4: Integration testing (dry run with 2 sessions)
- Phase 5: Hardening (max restart limits, PID file, log rotation)

> **WHY THIS MATTERS** Auto-start and manage all your Claude sessions from one command. No more manually opening 3 terminal windows.

MT-32 Visual Excellence & Design Engineering

In Progress

Phase 2/7

NEW (S118). Phase 1 starting — report_charts.py integration + research scan.

REMAINING

- Phase 3: Design system v2 (design tokens, lint rules, cross-format consistency)
- Phase 4: UI component library (reusable patterns for Claude Code web outputs)
- Phase 5: Dashboard v2 (interactive, real-time, responsive, themeable)
- Phase 6: Figure/image generation pipeline (multi-panel, annotation, export)
- Phase 7: Integration — all pillars feeding into reports, dashboards, slides, websites

> **WHY THIS MATTERS** Makes all visual output (reports, charts, dashboards) look professional. Design quality you'd expect from a real product.

MT-33 Strategic Intelligence Report

In Progress

Phase 0/6

Unknown

REMAINING

- Phase 1: Deep dive research — what metrics matter, what data is available, what charts...
- Phase 2: Data collection layer — SQL queries, data transformers, schema mapping
- Phase 3: Report section templates — Typst layouts for each pillar
- Phase 4: Chart integration — wire real data into chart_generator.py chart types
- Phase 5: Self-reference hook — JSON export, report diffing, chat-parseable format

MT-34 **Medical AI Tool — OpenEvidence Replacement** In Progress

IDEA LOGGED — Matthew needs more time to refine the concept. Do NOT start building until Matthew gives the green light.

Pending & Blocked

6 tasks awaiting start or unblocked

MT-5 **Claude Pro ↔ Claude Code Bridge** Not Started

Future task. Needs research on Claude Pro's current integration options.

MT-8 **iPhone Remote Control Perfection** Not Started

Not started. Needs r/ClaudeCode + r/ClaudeAI research sweep for remote control posts.

MT-18 **Academic Writing Workspace** Not Started

Not started. ClaudePrism repo: github.com/delibae/claude-prism. Session 39 created this task.

MT-19 **Local LLM Fine-Tuning Capability** Not Started

Not started. Long-term exploration. Unsloth repo: github.com/unslothai/unsloth.

MT-25 **Presentation Generator** Blocked

NOT STARTED. BLOCKED on style samples from Matthew.

MT-31 **Gemini Pro Integration** Not Started

Future task. Research phase needed. Low priority relative to 1-3 above.

WHY THIS MATTERS Use Google's Gemini Pro for visual/design tasks where it excels, while Claude handles the code. Best of both models.

PRIORITY QUEUE

Priority Queue

Active tasks ranked by decay-based priority scoring. Score = base value + aging penalty. Higher = work on this first.

1	MT-21	**Hivemind coordination** Phase 2 PASSED (6th). Phase 3 = 3-chat	10
2	MT-23	**Mobile Remote Control v2** Research: evaluate Telegram/Discord channels MCP	9
3	MT-26	**Financial Intelligence Engine** Research: scope trading/prediction market academic backbone	8
4	MT-28	**Self-Learning v2 (Multi-Domain)** Research: cross-domain YoYo, Sentinel adaptive mutation	8
5	MT-32	**Visual Excellence & Design Engineering** Phase 1: wire report_charts + nuclear scan. Absorbs MT-24/MT-25	8
5	MT-33	**Strategic Intelligence Report** Phase 1: deep dive research — data sources, metrics, chart mapping	10
—	MT-34	**Medical AI (OpenEvidence replacement)** BLOCKED: Matthew refining concept. Do not start.	6
7	MT-27	**CCA Nuclear v2** Research: APF improvement, better frontier tagging	5
8	MT-29	**Cowork + Pro Bridge Hivemind** Research: evaluate Cowork, bridge Pro↔Code	5
9	MT-18	**Academic writing** Research: install/evaluate ClaudePrism	7
10	MT-13	**iOS/macOS app dev** Phase 3: first real app	7

LIVE INFRASTRUCTURE

Live Infrastructure

18 hooks integrated into Claude Code via settings.local.json. Every tool call passes through this pipeline.

Event	Matcher	Hook	Purpose
PostToolUse	*	meter.py	Token counter
PostToolUse	*	compact_anchor.py	Context anchor writes
PostToolUse	*	senior_dev_hook.py	SATD + effort + quality scoring
PostToolUse	*	queue_hook.py	Cross-chat queue context
PreToolUse	*	alert.py	Warn/block at red/critical
PreToolUse	*	cost_alert.py	Cost threshold warning
PreToolUse	*	path_validator.py	Dangerous path detection
PreToolUse	*	edit_guard.py	Edit retry prevention
PreToolUse	*	validate.py	Spec guard + plan compliance
PreToolUse	Bash	credential_guard.py	Credential extraction guard
PreToolUse	Bash	bash_guard.py	Command safety (network, packages, system)
UserPromptSubmit	*	skill_activator.py	Spec auto-activation
UserPromptSubmit	*	skillbook_inject.py	Strategy injection
UserPromptSubmit	*	capture_hook.py	Real-time memory capture
UserPromptSubmit	*	queue_hook.py	Cross-chat queue injection
Stop	*	auto_handoff.py	Block exit at critical
Stop	*	capture_hook.py	Session-end memory capture
PostCompact	*	post_compact.py	Recovery + journal logging

INTELLIGENCE & RESEARCH

Intelligence & Research

Reddit Intelligence

FINDINGS BY VERDICT

BUILD	25
ADAPT	58
REFERENCE	237
PERSONAL	0
SKIP	48

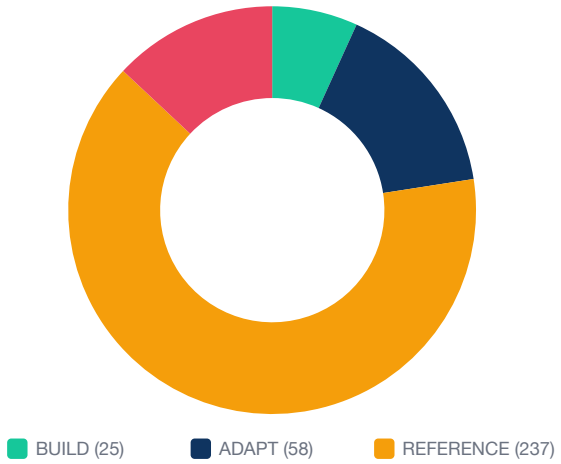
7 subreddits · 30 repos evaluated

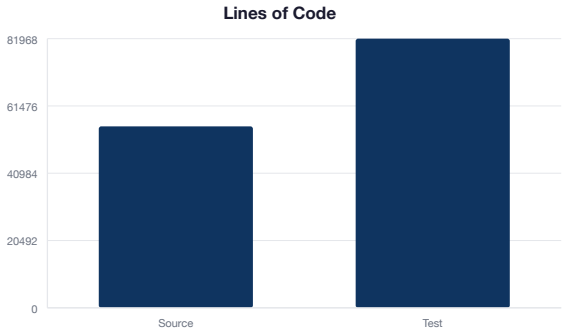
Self-Learning System

Strategies	10 total (4 confirmed)
Proposals	6 (all LOW risk)
Trace Sessions	31 analyzed
Avg Score	70 / 100
Papers Logged	**1242**
Sentinel Rate	5-10%

YoYo loop: observe → detect → hypothesize → build → validate

Intelligence Verdicts

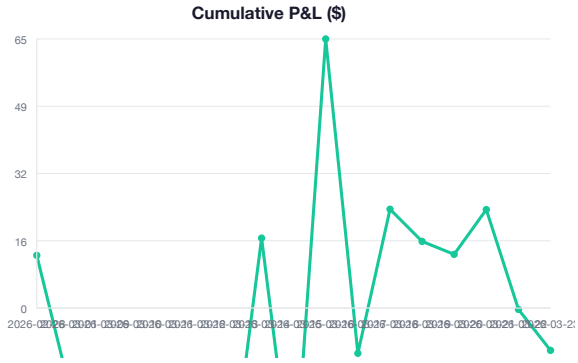


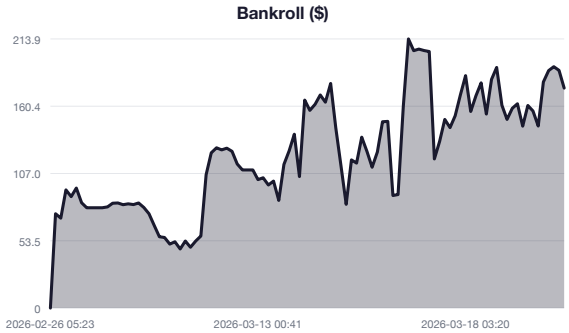


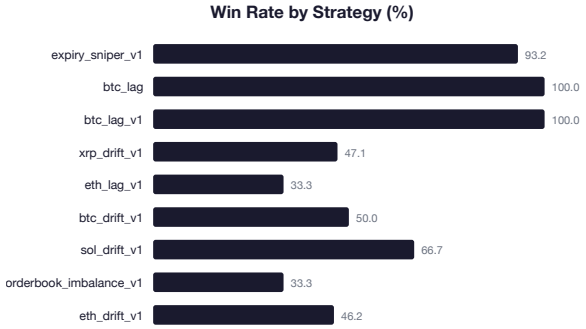
FINANCIAL ANALYTICS — KALSHI BOT

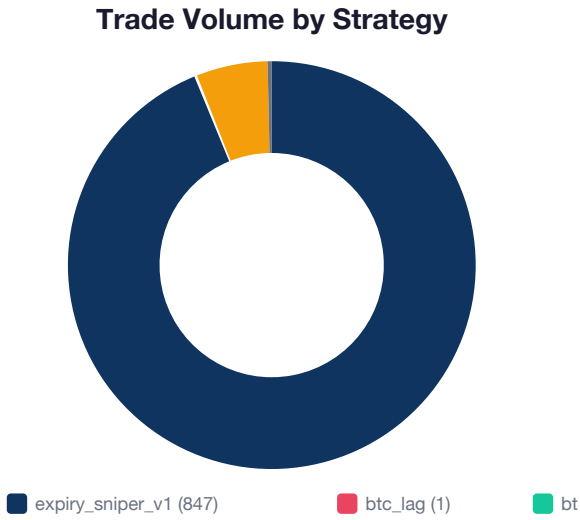
Financial Analytics — Kalshi Bot

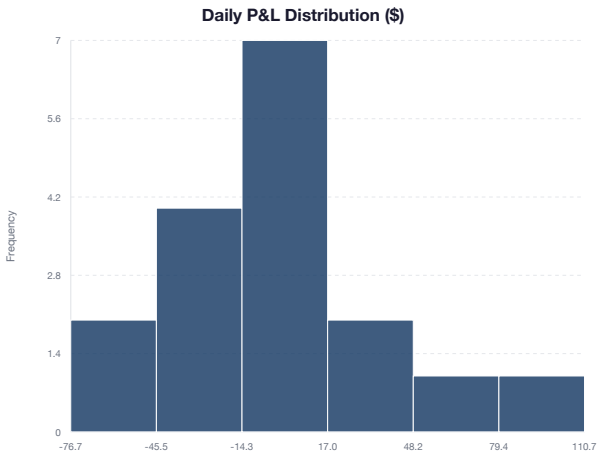
LIVE TRADES	WIN RATE	TOTAL P&L	SETTLED
1199	80.3%	$−10.33	1199

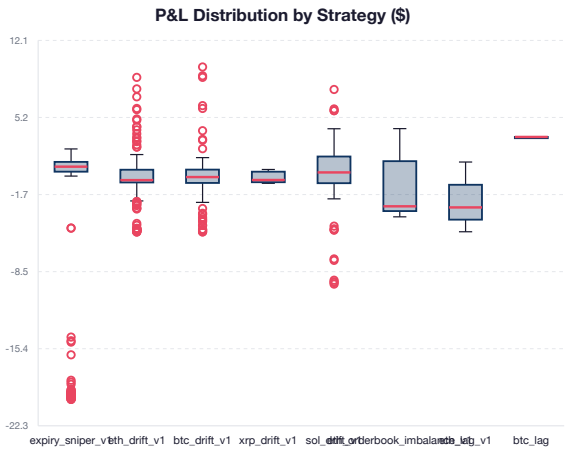












No data

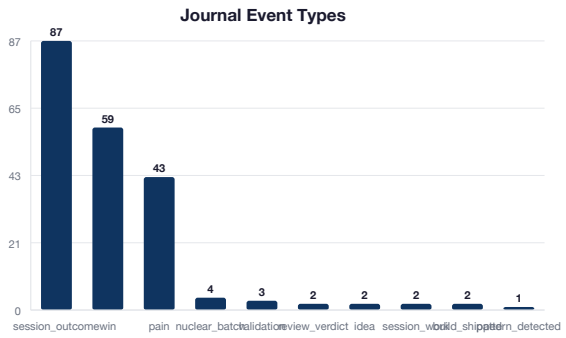
Strategy Performance

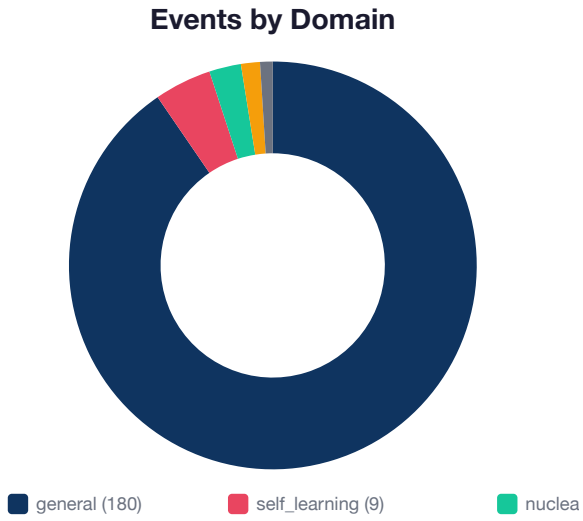
Strategy	Trades	Wins	Win %	Avg P&L	Total P&L
expiry_sniper_v1	847	789	93.2%	$0.08	**$65.25**
btc_lag	1	1	100%	$3.5	**$3.5**
btc_lag_v1	1	1	100%	$0.57	**$0.57**
xrp_drift_v1	51	24	47.1%	$−0.07	**$−3.8**
eth_lag_v1	3	1	33.3%	$−2.18	**$−6.53**
btc_drift_v1	78	39	50%	$−0.12	**$−9.69**
sol_drift_v1	45	30	66.7%	$−0.31	**$−14.08**
eth_orderbook_imbalance_v1	15	5	33.3%	$−1.21	**$−18.2**
eth_drift_v1	158	73	46.2%	$−0.17	**$−27.35**

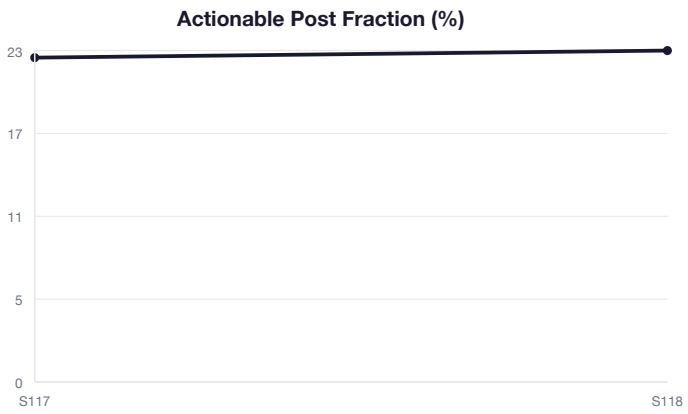
SELF-LEARNING INTELLIGENCE

Self-Learning Intelligence

JOURNAL ENTRIES	WINS / PAINS	CURRENT APF	POSTS REVIEWED
854	59 / 43	23.2%	340







ARCHITECTURE DECISIONS

Architecture Decisions

Decision	Rationale
Local-first storage	User owns all data, no cloud dependency
Stdlib-first (no pip)	Zero dependency management overhead
Hook-based delivery	Native Claude Code integration points
Adaptive context thresholds	Fixed percentages break at 1M windows
JSONL for all logging	Append-only, grep-friendly, no corruption risk
QualityGate geometric mean	Prevents Goodhart's Law gaming of metrics
Whitelist-first phishing detection	Zero false positives on legitimate domains
Typst over WeasyPrint	Single binary, millisecond compile, JSON-native

RISKS & BLOCKERS

Risks & Blockers

MT-1 Maestro Visual Grid UI

BLOCKER

Blocked on macOS 15.6 beta SDK crash. Tauri/React app crashes on launch.

Mitigation: Using tmux + dev-start script as workaround. Waiting for stable macOS release.

Semantic Scholar Rate Limiting

RISK

429 errors at 1.5s delay between queries for academic paper scanning.

Mitigation: Increased to 3s delay with exponential backoff.

Context Burn from File-Writing Hooks

DEBT

compact_anchor.py writes trigger system-reminder context consumption.

Mitigation: Writes limited to every 10 turns, files kept small.

NEXT PRIORITIES

Next Priorities

1. **Check SESSION_STATE.md for current priorities**

HONEST ASSESSMENT

Honest Assessment

Objective gaps, limitations, and areas where the project falls short of its stated goals. Included for accountability.

11 master tasks stalled at Phase 1

GAP

MT-0, MT-9, MT-11, MT-12, MT-13, MT-21, MT-22, MT-27, MT-28, MT-30, MT-33 — started but no subsequent phases shipped. Risk of scope sprawl without depth.

Single-developer project with no external users

LIMITATION

All tooling built for one developer. No community testing, no feedback loop from other Claude Code users. Limits validation of 'community-demanded' features.

High test count (5,735) may overstate coverage

NUANCE

Average 56 tests per source file. Many test trivial properties (dict keys, return types). Integration test coverage across modules is limited to 1 suite.

1 task(s) blocked with no resolution timeline

BLOCKER

MT-25: Presentation Generator

Self-learning metrics partially hardcoded

DEBT

Strategy count, proposal count, and avg score are hardcoded in the data collector rather than dynamically parsed from journal/strategy files.

Kalshi bot integration remains read-only

GAP

CCA scans for research and surfaces findings, but no closed-loop feedback exists: no tracking of which CCA recommendations led to profitable Kalshi trades.

CI/CD pipeline exists but unverified

NUANCE

GitHub Actions workflow defined but no evidence of runs in git history. All testing is local-only.

STATUS

All Systems Operational

7,850	137,206	122	754
TESTS	LOC	SESSIONS	COMMITS

9 modules · 33 master tasks · 198 test suites · Zero external dependencies

8	19	368
MTs Complete	MTs Active	Findings

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